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Aria-NBV: Quality-Driven Next-Best-View Planning with Egocentric Foundation Models

TODO: Deutscher Titel

Master's Thesis / Masterarbeit

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Transparency in the Use of AI Tools

TODO: Document AI tools used during thesis writing and implementation according to the final institutional requirements.

Acknowledgements

TODO: Acknowledgements.

Abstract

This thesis investigates quality-driven NBV planning for egocentric 3D reconstruction in ASE. It builds on the ARIA-NBV seminar paper and frames Relative Reconstruction Improvement (RRI) as the central supervision and evaluation signal for choosing informative future views.

Zusammenfassung

TODO: Deutsche Zusammenfassung ergaenzen.

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1 Introduction

Next-Best-View (NBV) planning selects future sensor views to improve a reconstruction under a finite acquisition budget. ARIA-NBV focuses this problem on egocentric indoor trajectories, where candidate views should be ranked by expected reconstruction quality instead of by geometry-only coverage proxies.

The thesis starts from the seminar-paper formulation of RRI{}-driven NBV and extends it toward a thesis-scale treatment of quality-driven view selection, model calibration, and planning behavior.

2 Background

The project combines ASE scene assets, EFM3D features, oracle RRI labels, and VIN-style candidate scoring. Shared notation and glossary definitions are imported from the ARIA-NBV Typst shared modules, so thesis chapters can use symbols such as RRI and equations such as

$$\text{RRI}(q) = \frac{\text{CD}(\mathcal{P}_t, \mathcal{M}_{\text{GT}}) - \text{CD}(\mathcal{P}_t \cup \mathcal{P}_q, \mathcal{M}_{\text{GT}})}{\text{CD}(\mathcal{P}_t, \mathcal{M}_{\text{GT}}) + \varepsilon}$$

.

3 Method

TODO: Describe the final thesis method, including the oracle labeler, candidate generation, VIN scorer, and any non-myopic planning extension that remains in scope.

4 Evaluation

TODO: Define the thesis evaluation protocol, datasets, metrics, and ablations. Keep cumulative target reconstruction quality and acquisition cost visible before introducing scalarized objectives.

5 Conclusion

TODO: Summarize thesis contributions, limitations, and follow-up work.

List of Figures

Appendix A: Supplementary Material

TODO: Add supplementary derivations, additional tables, and implementation details.

Bibliography

Ehrenwoertliche Erklaerung

Ich versichere, dass ich diese Masterarbeit selbststaendig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Muenchen, TODO: Abgabedatum

Jan Duchscherer